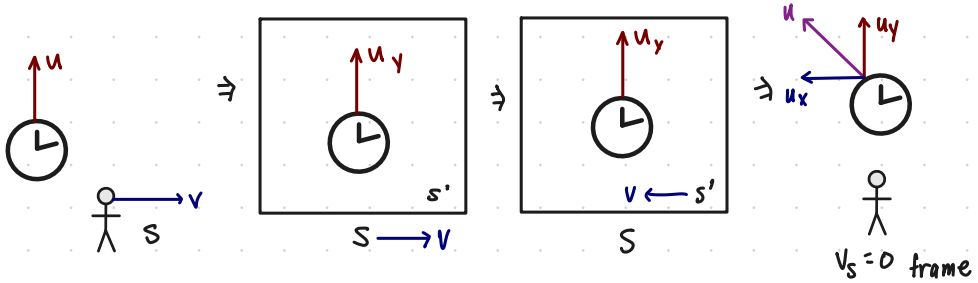
2.19. **Slanted time dilation** *

A clock moves vertically with speed u in a given frame, and you run horizontally with speed v with respect to this frame. Show that you see the clock run slow by the nice simple factor $\gamma_u \gamma_v$.



from your frame, the components of your horizontal velocity are added to the vertical velocity of the clock. Since c is relative, the addition of these components must be transformed by the lorentz factor to not exceed this value.

proof:

from S' the lorentz transform yields:

$$\gamma_u = \frac{1}{\sqrt{1 - u^2/c^2}}$$

we find $dt' = \gamma_u d\tau$

from S the lorentz transformation yields:

$$dt' = \gamma_v \left(d\tau - v\,dx/c^2 \right)$$

where $dx = 0$, thus:

$$dt = \gamma_v \, dt'$$

putting these together by subbing dt' in S

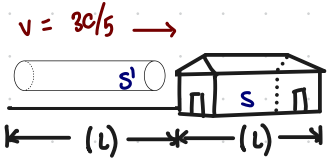
gives us $dt = \gamma_u \gamma_v \, d\tau$

and resolves $\frac{dt}{d\tau} = \gamma_u \gamma_v$

2.25. **Pole in barn***

(a) A pole with proper length L moves rightward at speed $3c/5$ through a barn with proper length L. In the barn frame, what are the Δx and Δt separations between the "left end of pole passing left end of barn" and "right end of pole passing right end of barn" events?

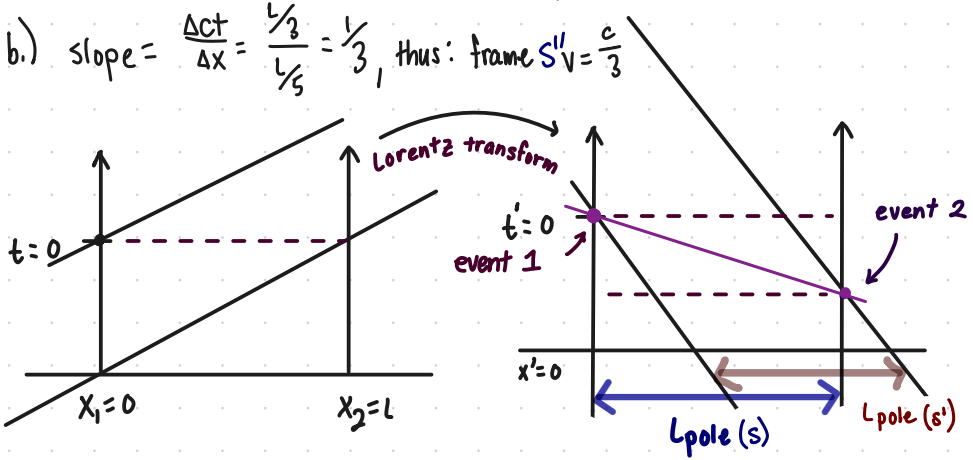
(b) If you analyze the setup in the pole frame, you will quickly see that the order of the above events is reversed from the order in the barn frame. By continuity, there should therefore be a frame in which the two events happen at the same time. Use a Minkowski diagram to find this frame.



$$\gamma = \frac{1}{\sqrt{1-v^2/c^2}} = \frac{1}{\sqrt{1-(3c/5)^2/c^2}} = 5/4$$

$$S: \Delta x = L - \frac{L}{\gamma} = L - \frac{L}{5/4} = L/5 \qquad \Delta t = \frac{\Delta x}{v} = \frac{L/5}{3c/5} = L/3c$$

b.) $\text{slope} = \frac{\Delta ct}{\Delta x} = \frac{L/3}{L/5} = 1/3$, thus: frame S'' $v = \frac{c}{3}$



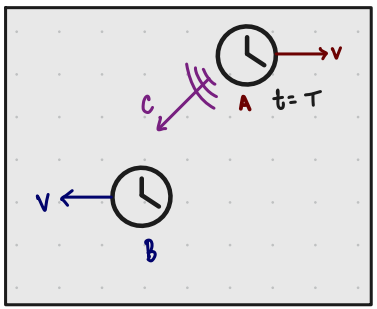
1.24. **Simultaneous waves***

Alice flies past Bob at speed v. Right when she passes, they both set their watches to zero. When Alice's watch shows a time T, she waves to Bob. Bob then waves to Alice simultaneously (as measured by him) with Alice's wave (so this is before he actually *sees* her wave). Alice then waves to Bob simultaneously (as measured by her) with Bob's wave. Bob then waves to Alice simultaneously (as measured by him) with Alice's second wave. And so on. What are the readings on Alice's watch for all the times she waves? And likewise for Bob?

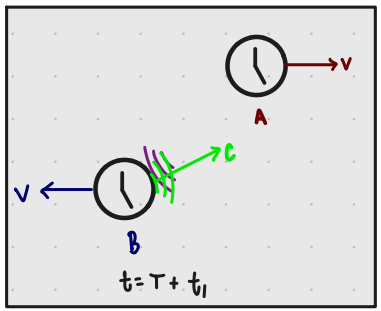
2.28. **Simultaneous waves again****

Solve Exercise 1.24 by using a Minkowski diagram from the point of view of an observer who sees Alice and Bob moving with equal and opposite speeds.

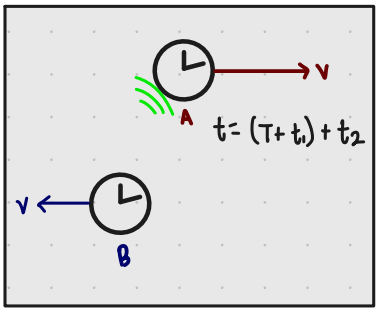
$V = V$



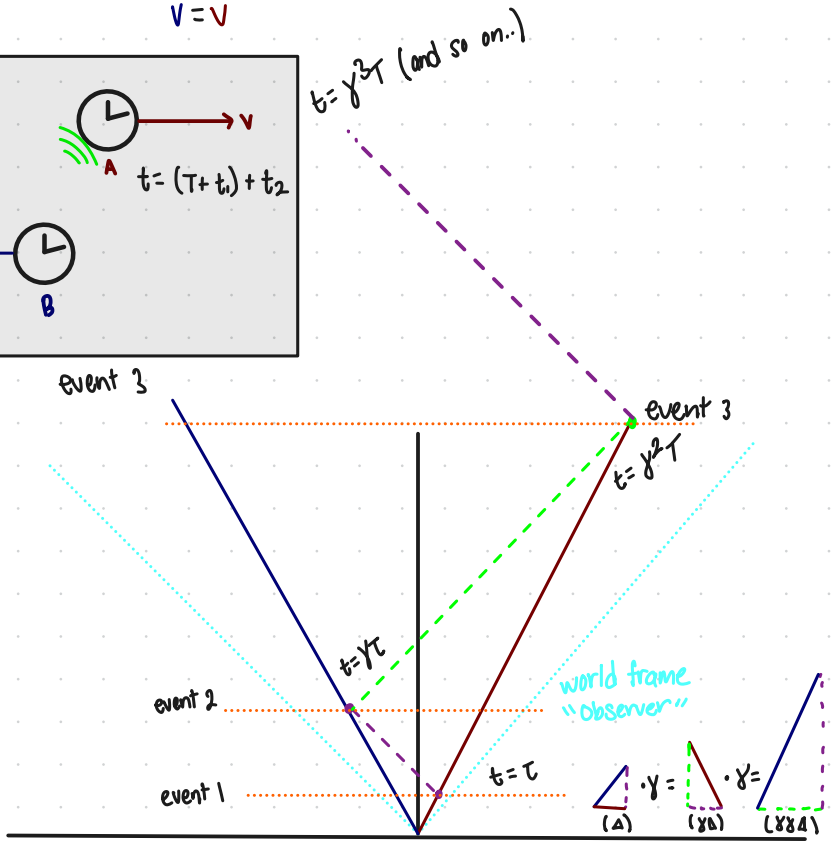
event 1



event 2



event 3



2.29. **Short train in a tunnel again*****

Solve Exercise 1.39 by using a Minkowski diagram from the point of view of the train, and also of the tunnel.

1.39. **Short train in a tunnel ****

Consider the scenario in Problem 1.7, with the only change being that the train now has length rL, where r is some numerical factor. What is the largest value of r, in terms of v, for which it is possible for the bomb not to explode? Verify that you obtain the same answer working in the train frame and in the tunnel frame.

1.7. **Train in a tunnel ****

A train and a tunnel both have proper length L. The train moves toward the tunnel at speed v. A bomb is located at the front of the train. The bomb is designed to explode when the front of the train passes the far end of the tunnel. A deactivation sensor is located at the back of the train. When the back of the train passes the near end of the tunnel, the sensor sends a signal to the bomb, telling it to disarm itself. Does the bomb explode?

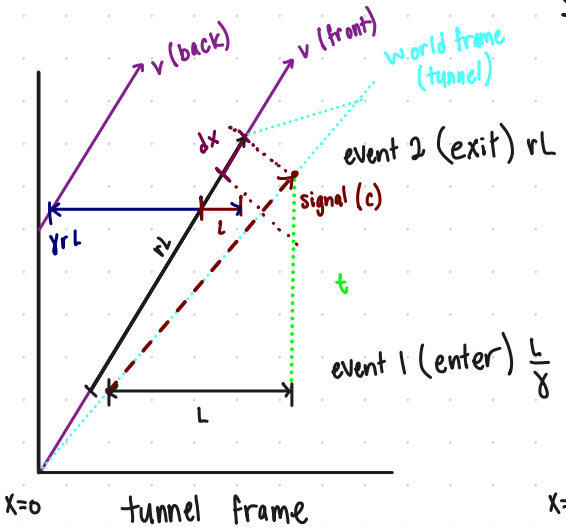
* v dictates Ø of train in tunnel frame, Ø dictates max r of rL

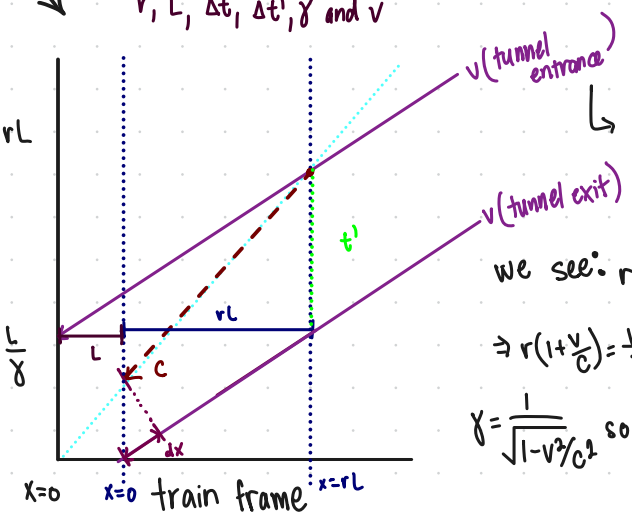
as we can see, in order to resolve the max value of r we need to set the equality of rL in terms of r, L, Δt, Δt', γ and v

we find:

$t = \frac{rL}{c}$ (signal) and $rL = \frac{L}{\gamma} - v \cdot t'$ (exit)

↳ when $t = t' = \tau$: $rL = \frac{L}{\gamma} - vt$ and $t = \frac{rL}{c}$





we see: $rL = \frac{L}{\gamma} - v\left(\frac{rL}{c}\right)$, solving for r we get. $r = \frac{1}{\gamma} - v\frac{r}{c} \Rightarrow r + vr/c = \frac{1}{\gamma}$

$\Rightarrow r\left(1+\frac{v}{c}\right) = \frac{1}{\gamma} \Rightarrow r = \frac{1}{\gamma(1+v/c)}$

$\gamma = \frac{1}{\sqrt{1-v^2/c^2}}$ so $r = \sqrt{\frac{1-v^2/c^2}{1+v^2/c^2}} = \sqrt{\frac{1-v/c}{1+v/c}}$

2.30. **Running toward each other ****

A and *B* move toward each other. They each have speed *v* with respect to the ground. *A* sends out flashes at frequency *f* (as measured by her). With what frequency does *B* receive the signals (as measured by him)? Solve this in two ways:

(a) Imagine a stationary intermediate person *C* who receives the flashes and then immediately emits flashes. So the flashes effectively pass right through her. See Fig. 2.31(a).

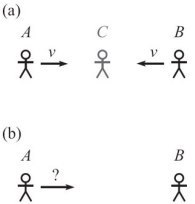


Figure 2.31

(b) Work in *B*'s frame, where *A* approaches at a certain speed. See Fig. 2.31(b).

a.)

A frame:

$$\Delta x' = \gamma(\Delta x - v\Delta t)$$

$$\Delta t' = \gamma\left(\Delta t - \frac{v}{c^2}\Delta x\right)$$

C frame:

$$\Delta x = \gamma(\Delta x + v\Delta t')$$

$$\Delta t = \gamma\left(\Delta t' + \frac{v}{c^2}\Delta x'\right)$$

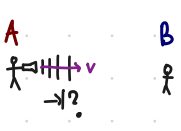
doppler effect: $\Delta T = \frac{1}{c}(c-v)\gamma t' = \frac{1-\beta}{\sqrt{1-\beta^2}}\Delta t'$

$$= \sqrt{\frac{1-\beta}{1+\beta}}\Delta t' = \sqrt{\frac{1-\beta}{1+\beta}} \cdot \frac{1}{f'} = f'\sqrt{\frac{1+\beta}{1-\beta}}$$

$\Rightarrow$ hence, $f_C = f_A\sqrt{\frac{1+v/c}{1-v/c}}$ and by substituting f_C for f_A in f_B in..

$$\therefore f_B = f_A\sqrt{\frac{1+v/c^2}{1-v/c^2}}, \text{ we see: } f_B = \left(f_A\sqrt{\frac{1+v/c}{1-v/c}}\right)\left(\sqrt{\frac{1+v/c}{1+v/c}}\right) = f_A\left(\frac{1+v/c}{1-v/c}\right)$$

b.)



$$\underset{u*}{A(v)} = \frac{v+v}{1+\frac{uv}{c^2}} = \frac{2v}{1+\frac{v^2}{c^2}} \Rightarrow f_A\left(\frac{1+u/c}{1-u/c}\right)$$

since $u = v*$

sub u into f_A:

numerator:

$$1 + \frac{2v/c}{1+v^2/c^2} \Rightarrow \frac{1+v^2/c^2 + \frac{2v}{c}}{1+v^2/c^2} = \left[\frac{\left(1+\frac{v}{c}\right)^2}{1+v^2/c^2}\right]$$

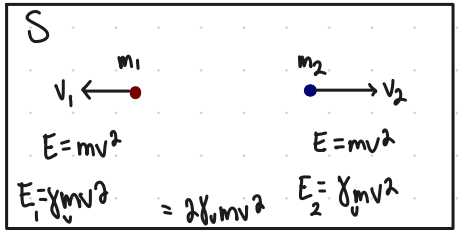
factor $\left(1+\frac{v}{c}\right)^2$

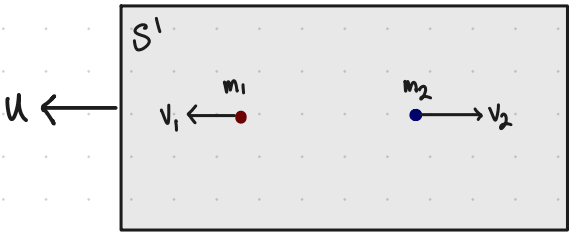
denominator: ... same process: $\left[\frac{\left(1+v/c\right)^2}{1-v^2/c^2}\right]$

$$\Rightarrow f_B = f_A\sqrt{\left(\frac{\text{num.}}{\text{den.}}\right)} = f_A\sqrt{\frac{\left(1+\frac{v}{c}\right)^2}{\left(1-v/c\right)^2}} = f_A\frac{\left(1+v/c\right)}{\left(1-v/c\right)}$$

3.24. **Energy of two masses ***

Two masses m move at speed v, one to the east and one to the west. What is the total energy of the system? Now consider the setup as viewed in a frame moving to the west at speed u. Find the energy of each mass in this frame. Is the total energy larger or smaller than the total energy in the lab frame?





speed of moving frame S'

$$u = \frac{v + u'}{1 + \frac{vu'}{c^2}}$$

speed of object measured in moving frame S'

speed of object measured in rest frame S

$$\hookrightarrow v_1' = \frac{\Delta x}{\Delta t}, \quad \left.\begin{aligned} \Delta x &= \gamma_u(\Delta x' - u\Delta t') \\ \Delta t &= \gamma_u(\Delta t' - u\Delta x'/c^2) \end{aligned}\right\} = v_1' = \frac{u - v}{1 - uv/c^2}$$

$$v_2' = \frac{\Delta x}{\Delta t}, \quad \begin{aligned} \Delta x &= \gamma_u(\Delta x' + u\Delta t') \\ \Delta t &= \gamma_u(\Delta t' + u\Delta x'/c^2) \end{aligned} \quad = v_2' = \frac{u + v}{1 + uv/c^2}$$

w/ v_1, v_2 @ S' resolved we find the energy w/

Gamma Identity: $\gamma(v') = \gamma_u \gamma_v \left(1 \pm \frac{uv}{c^2}\right)$ such that $E_1 = \gamma_u \gamma_v mc^2\left(1 - \frac{uv}{c^2}\right)$ & $E_2 = \gamma_u \gamma_v mc^2\left(1 + \frac{uv}{c^2}\right)$

so, $E_{TOT} = \gamma_u \gamma_v mv^2$

3.25. **CM frame ****

A mass m travels at speed $3c/5$, and another mass m sits at rest.

(a) Find the energy and momentum of each particle in the lab frame.

(b) Find the speed of the CM of the system, by using a velocity-addition argument, and then again by using the result from Problem 3.2.

(c) Find the energy and momentum of each particle in the CM frame, without using the Lorentz transformations.

(d) Verify that the E's and p's are related by the relevant Lorentz transformations.

(e) Verify that $E^2 - p^2c^2$ for each mass is the same in both frames. Likewise for $E_{\text{total}}^2 - p_{\text{total}}^2c^2$.

a.)